

Interview Trainer Agent

Problem Statement 22: Interview Trainer Agent

Objective

The objective of this project is to help students and job seekers prepare effectively for technical interviews using Artificial Intelligence.

The system analyzes a candidate's resume and generates personalized interview questions based on their skills, projects, technologies, and target job role. This provides focused interview preparation and improves confidence before attending real interviews.

Features

Resume Upload

Users can upload their resume in PDF or TXT format.

Role-Based Interview Preparation

Users can select a target role such as:

- Python Developer
- Java Developer
- Web Developer
- Data Analyst
- AI Engineer

Personalized Question Generation

The system generates interview questions based on the candidate's resume and selected role.

Difficulty Levels

Supports:

- Beginner
- Intermediate
- Advanced

AI-Powered Interview Preparation

Uses IBM Watsonx AI foundation models to generate intelligent and relevant interview questions.

User-Friendly Interface

Built with Streamlit for an interactive and easy-to-use experience.

Technology Stack

Frontend

- Streamlit

Programming Language

- Python

AI Platform

- IBM Watsonx AI

Agent Framework

- LangFlow

Cloud Platform

- IBM Cloud

Libraries Used

- ibm-watsonx-ai
 - pypdf
 - streamlit
 - langflow
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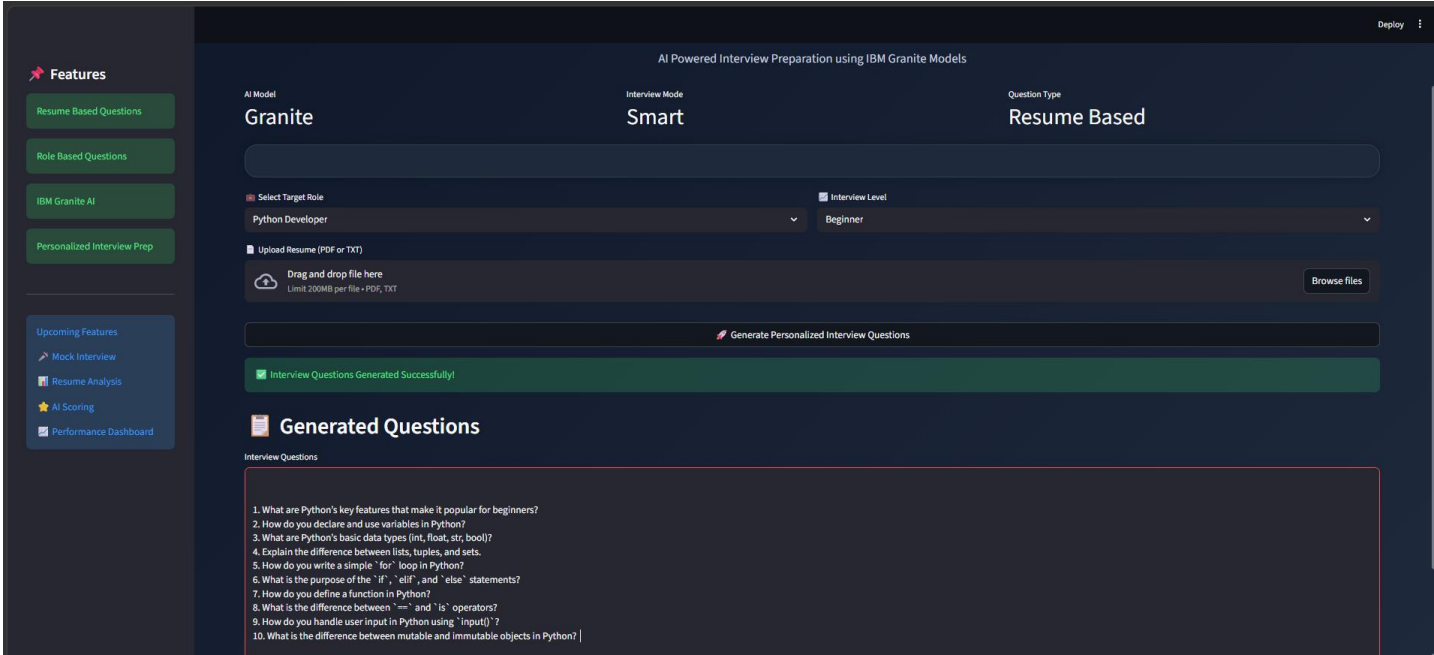
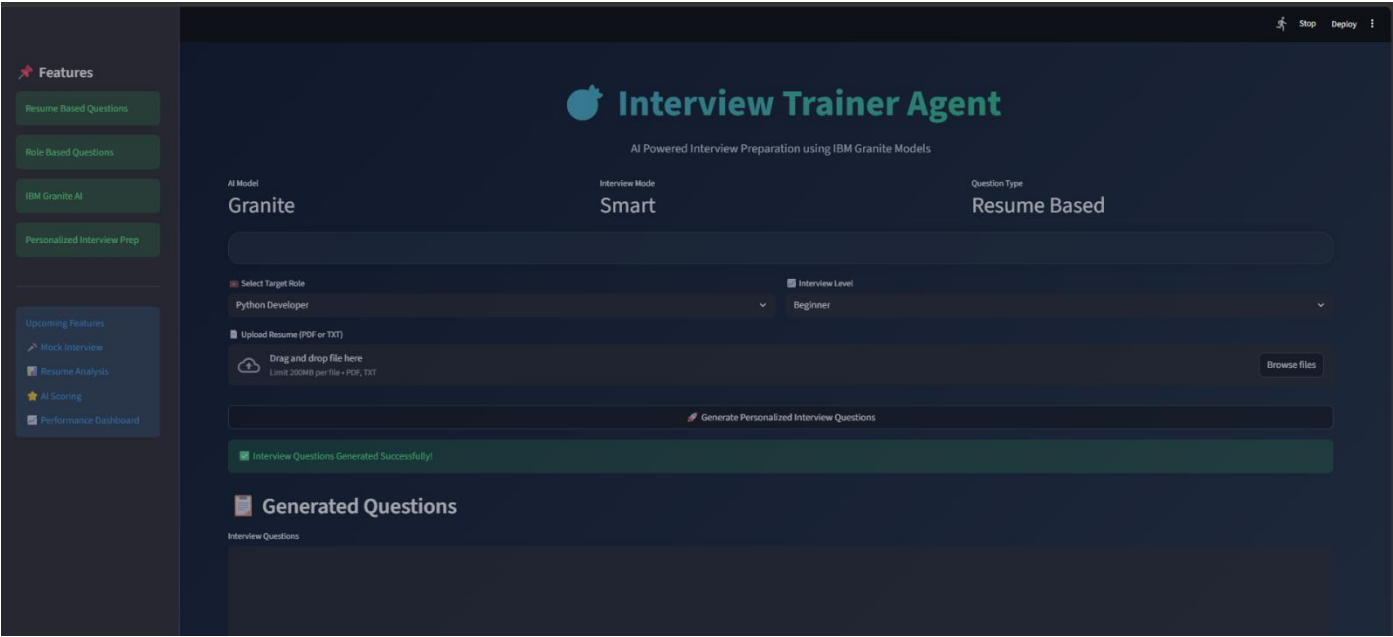
Expected Outcome

The application generates personalized interview questions based on a user's resume and selected job role, helping candidates improve their interview preparation and readiness.

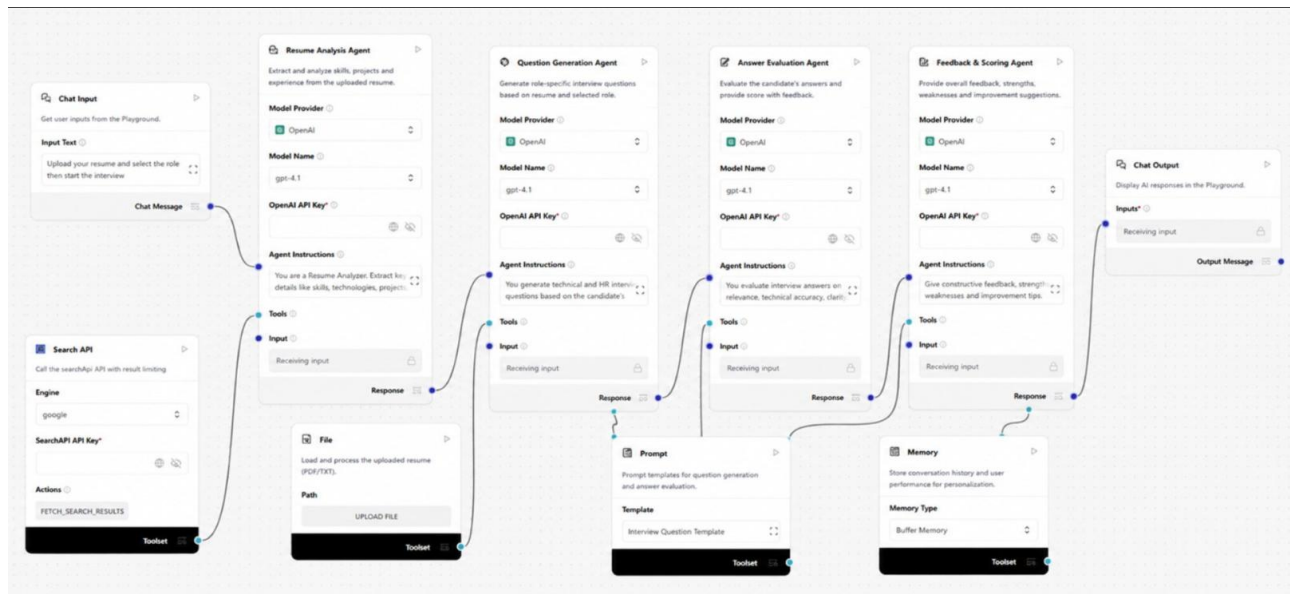
Screenshots:

The screenshot displays the IBM watsonx console interface. At the top, the header bar includes the IBM watsonx logo, an 'Upgrade' button, a help icon, a notification bell, the user account 'Sai sumanth's Account', the location 'Dallas', and a profile icon. Below the header, a breadcrumb trail shows 'Projects / Sai's sandbox'. A navigation bar contains tabs for 'Overview', 'Assets', 'Deployments', 'Jobs', and 'Manage', with 'Manage' being the active tab. On the left, a sidebar lists project management options: 'General' (selected), 'Access control', 'Environments', 'Resource usage', 'Services & integrations', 'Tools', and 'Pipeline'. The main content area is divided into two panels. The 'General' panel on the left shows project details: Name ('Sai's sandbox'), Description ('A project to try things in'), Tags (with a note to 'Add tags to make projects easier to find.'), and Project ID ('89588250-15c0-48b9-92ff-19e7535f2130'). The 'Storage' panel on the right shows 'Storage used' as '0 Bytes', the 'Bucket' name 'saissandbox-donotdelete-pr-6662jr9580e0xh', and a 'Regenerate credentials' button with a note 'Last updated on: Jun 04, 2026, 09:45 PM'. A 'Manage in IBM Cloud' link is also present.

The screenshot shows a code editor with a dark theme. The Explorer sidebar on the left lists the project files: 'InterviewTrainerAgent', '__pycache__', 'config.python-313.pyc', 'app.json', 'app.py', 'config.py', 'Readme.md' (selected), and 'requirements.txt'. The main editor area displays the content of 'Readme.md', which is a markdown file titled '# Interview Trainer Agent using IBM Watsonx AI'. The file includes sections for 'Problem Statement', 'Project Description', 'Features', and 'Technology Stack'. The 'Features' section lists: Resume Upload (PDF/TXT), Personalized Interview Questions, Role-Based Question Generation, Beginner, Intermediate and Advanced Levels, IBM Watsonx AI Integration, LangFlow Workflow, and Interactive Streamlit Interface. The 'Technology Stack' section lists: Python, Streamlit, IBM Watsonx AI, LangFlow, IBM Cloud, and PDF Processing.



Langflow Process



Blueprint

